FINAL YEAR PROJECT PROPOSAL

DECENTRALIZED ELECTRONIC HEALTH RECORD (EHR) MANAGEMENT SYSTEM VIA SMART CONTRACTS

INU

Submitted By:

Bilal Hassan (20112)

Hamza Ali (20115)

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Supervised By:

Engr. Sana Fatima

A project proposal submitted to The IQRA National University, Peshawar in partial fulfillment of the requirements for the degree of

BACHELOR OF SCIENCE (BS) IN COMPUTER SCIENCE

SCHOOL OF COMPUTER SCIENCE

IQRA NATIONAL UNIVERSITY, PESHAWAR-PAKISTAN

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List of Acronyms

Acronym	Description
INU	IQRA National University
AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
API	Application Programming Interface
UI / UX	User Interface / User Experience
BLE / IoT	Bluetooth Low Energy / Internet of Things

1. Introduction & Problem Statement

The project titled **"Decentralized Electronic Health Record (EHR) Management System via Smart Contracts"** addresses critical architectural paradigms within the modern computer science landscape. As information engineering systems shift toward distributed, agile environments, specialized automated monitoring and optimization models become absolutely paramount to unlock next-generation performance.

Problem Statement: Modern healthcare setups house patient records within rigid siloed databases susceptible to malicious single-point-of-failure breaches and illegal alterations. Inter-hospital record sharing remains slow, highly fragmented, and heavily exposed to systemic credential leaks, creating massive friction when immediate validation is essential during unexpected medical emergencies. Without

an automated, reliable system structure to streamline these processes, operational risks escalate dramatically. This system solves this missing gap by developing an optimized computational pipeline that acts as a robust mechanism for resolving these complex challenges.

2. Objectives of Project

The primary structural milestones mapped for this software development cycle are:

- Architect a highly resilient, permissioned consortium blockchain network layout utilizing Hyperledger Fabric framework primitives.
- Formulate secure execution smart contracts enforcing strict patient-governed access delegation controls over granular medical data items.
- Build an intuitive cross-platform administrative portal enabling certified medical professionals to retrieve records upon validation safely.

3. Methodology & System Architecture

The system adheres to a structured, modular pipeline layout designed using a decoupled Client-Server model. This ensures that heavy backend model inferencing jobs do not impair the frontend interface application execution parameters.

Core Modules:

1. **Authentication & Profile Integration Module:** Handles user record enrollment and data secure caching sequences smoothly.
2. **Core Execution and Inference Pipeline:** Manages real-time data ingestion, transformation operations, and algorithmic processing nodes.
3. **Analytics & Reporting Layout Engine:** Aggregates calculation products, compiling structured system reports safely.

[System Architecture Design: React Native Client Interface ←→ Secure Gateway APIs ←→ Python/ Node Compute Processing Engines ←→ Scalable Vector/Relational Databases]

4. Proposed Time Schedule

Phase	Activity Description	Duration (2025-2026)
Phase 1	Literature Survey & Component Finalization	December
Phase 2	Architecture Modeling & Engineering Checks	January - February
Phase 3	Core Engine Development & Coding Cycle	March - April
Phase 4	System Verification Testing & Report Synthesis	May - June

Visual Timeline (Gantt Chart Representation)

Activity	Dec	Jan	Feb	Mar	Apr	May
Literature Gathering	■					
Scheme Design Modeling		■	■			
System Core Coding				■	■	
Verification & Writing						■

5. Budget Description & Technical Stack

The project leverages open-source libraries and public cloud deployment tiers, achieving maximum software efficiency with minimal capital expenditures. All computation and model runs are hosted on standard developer hardware platforms already in position.

Proposed Technical Core Stack: React Native, Node.js REST API, Hyperledger Fabric, IPFS (InterPlanetary File System), Express, Cryptography.js.

6. Undertaking & Project Declarations

We hereby declare that this technical development proposal is our genuine work executed under supervision at IQRA National University. No portion of this engineering research layout has been submitted previously for academic credit at any alternative educational board.

Student 1 Signature

Student 2 Signature

7. Institutional Review Panel Logs

Supervisor Evaluation Section:

_____

Supervisor Signature Date: ____/____/_____

FYP Department Coordinator Workspace:

_____

Coordinator Signature Status: APPROVED / HOLD